

REGIONAL MARITIME UNIVERSITY

MARINE ELECTRICAL/ELECTRONICS ENGINEERING

BCE/BEE/BMT/BCE/BME/ 27B

END OF SEMESTER EXAMINATION

BEE 207 – BASIC ELECTRICITY

(Applied Electricity)

INSTRUCTIONS: ANSWER FOUR QUESTIONS IN ALL. Each carry equal (25) marks. TIME ALLOWED: 3HRS

1. (a) Fig 1.0 shows a circuit containing two sources of e.m.f., each with their internal resistance. Determine the current in each branch of the network by using the superposition theorem.

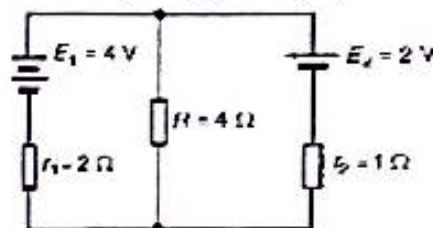


Fig 1.0

- (b) An 8H inductor has a current of 3A flowing through it. How much energy is stored in the magnetic field of the inductor?

2. (a) The maximum values for the sinusoidal alternating voltage and current are 400 V and 20 A respectively in a circuit connected to 50 Hz supply. The instantaneous values of the voltage and current; 285 V and 16 A respectively at $t = 0$ (both increase positively)

- Write down the expression for voltage and current at time, t .
- Determine the power consumed in the circuit.

- (b) George Simon Ohms is a German physicist who brought about the Ohm's Law. Kindly state his Law and give its mathematical expression.

3. (a) Find the current passing through the 3Ω resistor using Kirchhof's Laws in Fig 2.0.

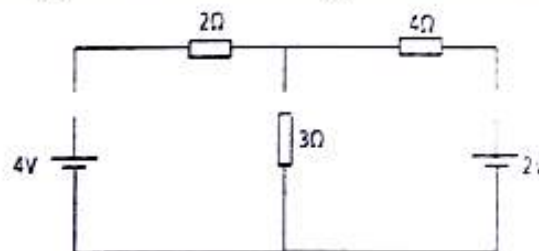


Fig 2.0

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Doc. No.: EXM/F/03	Page No.: 1/1	Issue No.: 1	Last Update: 15/04/15
Revision No.: 1	Date: 12/04/06	Author: EC	Approved: PROVOST

(b) A conductor 300mm long moves at a uniform speed of 4 m/s at right-angles to a uniform magnetic field of flux density 1.25 T. Determine the current flowing in the conductor when;

- (i) its ends are open-circuited,
- (ii) its ends are connected to a load of 20 Ohm resistance.

4. (a) A mild steel ring has a radius of 50mm and a cross-sectional area of 400mm². A current of 0.5A flows in a coil wound uniformly around the ring and the flux produced is 0.1mWb. If the relative permeability at this value of current is

200, find;

- (i) the reluctance of the mild steel and
- (ii) the number of turns on the coil.

(b)) Use Thevenin's theorem to find the current flowing in the 10Ω resistor for the circuit shown in Fig 3.0.

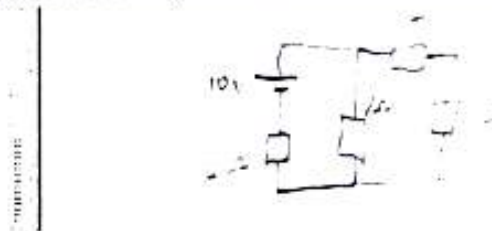


Fig 3.0

5. (a) Considering pure capacitance in an ac circuit element, show that the alternating current leads the voltage by $\pi/2$.

(b) Explain the principles of Electromagnetic Induction.